Memo - Straight lines Worksheet "Revision of Grade 9 Functions" Grade 10

Question 1

1.1 $\frac{y}{x} : \frac{5}{1} \neq \frac{7}{2} \neq \frac{10}{3}$

The x and y values are not in direct proportion so there is not a constant ratio (gradient)

1.2.1 $m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-8-8}{2-(-2)} = \frac{-16}{4} = -4$

using points $(2;-8)$ and $(-2;8)$

1.2.2 $(a:-16)$ & $(2:-8)$ & $(7:b)$

$m = \frac{-16-(-8)}{a-2} = -4$

$\frac{-8}{a-2} = -4$

$-8 = -4(a-2)$

$-8 = -4a + 8$

$4a = 16$

$a = 4$

$m = \frac{-8-b}{2-7} = \frac{-4}{1}$

$\frac{-8-b}{-5} = -4$

$-8-b = 20$

$-28 = b$

Question 2

2.1 $y = -\frac{1}{2}x - 2$ Sub $x = -4$ to see if $y = -4$

$y = -\frac{1}{2}(-4) - 2$

$y = 2 - 2$

$y = 0$.

$\therefore$ The point will not lie on the line as $y \neq -4$

2.2 $y = -\frac{1}{2}x - 2$ sub $(p:-6)$

$-6 = -\frac{1}{2}(p) - 2$

$\frac{1}{2}p = 4$

$p = 8$

Powered by CamScanner

2.3 $4y + 8x = -8$

$4y = -8x - 8$

$y = -2x - 2$

$\therefore m_1 = -2$

$y = -\frac{1}{2}x - 2$

$m_2 = -\frac{1}{2}$

$m_1 \times m_2$

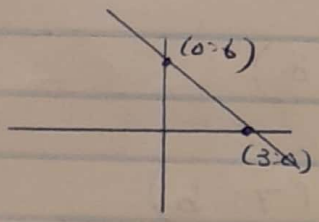
$= -2 \times -1/2$

$= 1$

$\therefore$ The lines are not perpendicular because $m_1 \times m_2 \neq -1$

Question 3

3.1



$y = mx + c$

$y = mx + 6$ sub (3;0)

$0 = m(3) + 6$

$-6 = 3m$

$-2 = m$

$\therefore y = -2x + 6$

Gradient $= \frac{\text{Rise}}{\text{Run}}$

$= \frac{\downarrow 6}{\rightarrow 3}$

$= \frac{-6}{3}$

$= -2$

3.2 $y = mx + c$ is // $y = -2x + 6$ $\therefore m = -2$

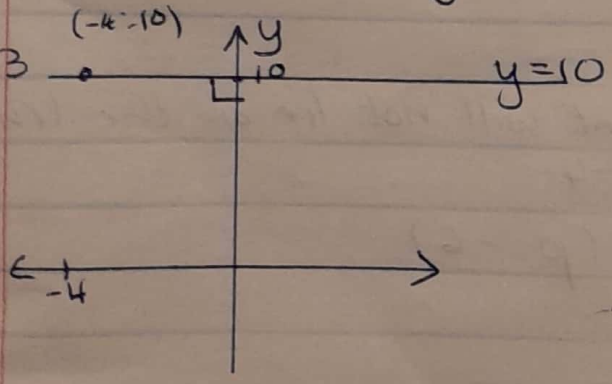
$y = -2x + c$ sub (-4;10)

$10 = -2(-4) + c$

$10 = 8 + c$

$2 = c$ $\therefore y = -2x + 2$

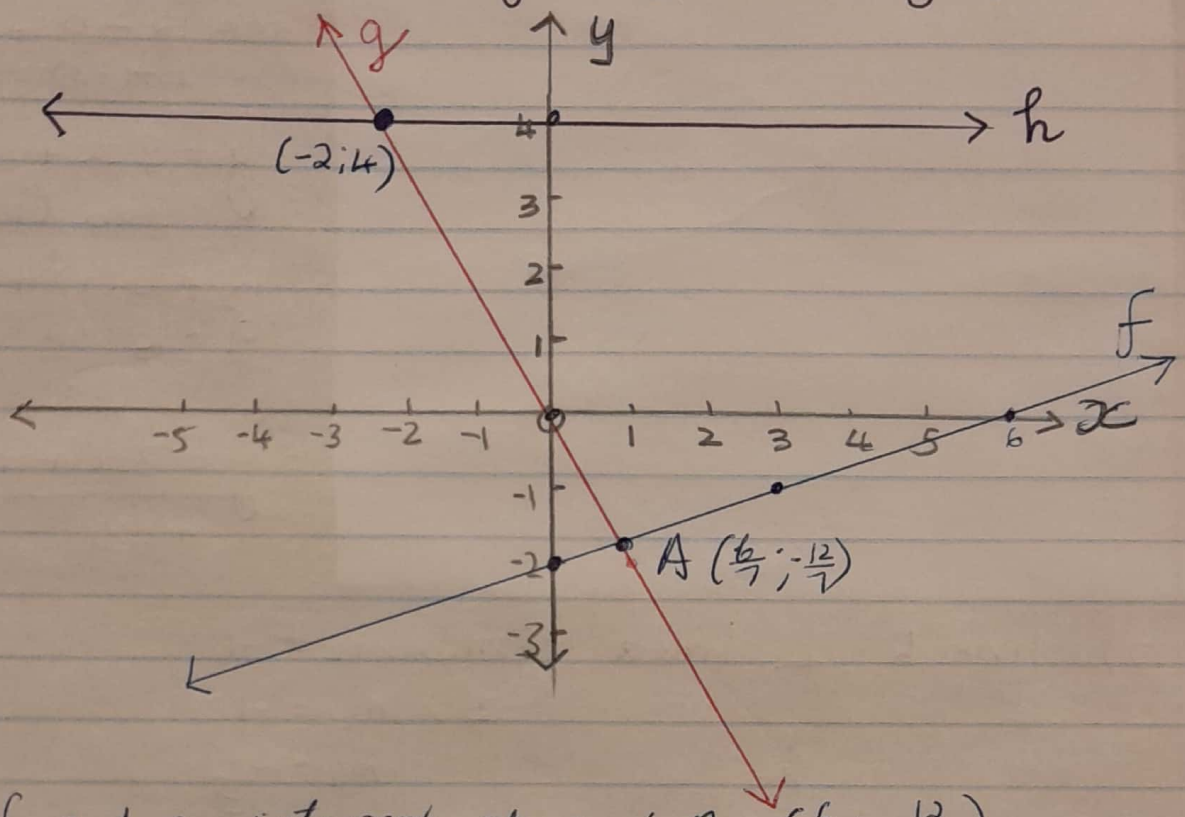
3.3



Powered by CamScanner

Question 4

4.1 $y = \frac{1}{3}x - 2$ Graph f $y = -2x + 0$ graph g $y = 4$ graph h



4.2.1 f and g intercept at point A $(\frac{6}{7}; -\frac{12}{7})$

Algebraically:

$y = \frac{1}{3}x - 2 \rightarrow ①$ and $y = -2x \rightarrow ②$

$-2x = \frac{1}{3}x - 2$

$2 = 2\frac{1}{3}x$

$\frac{6}{7} = x$

$y = -2\left(\frac{6}{7}\right)$

$y = -\frac{12}{7}$

4.2.2 $y = -2x \rightarrow ①$ and $y = 4 \rightarrow ②$

$4 = -2x$

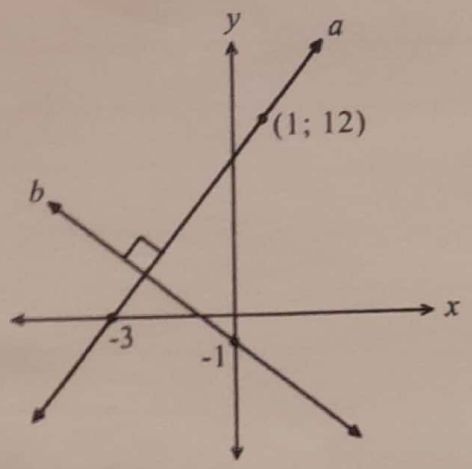
$-2 = x$ $\therefore y = 4$

Pt of intersection is $(-2; 4)$

Powered by CamScanner

Question 5

5.1



the information in the diagram to find the following:

1 the equation of line *a* (4)

2 the equation of line *b* (2)

1) line a

$m = \frac{12-0}{1-(-3)} = \frac{12}{4} = 3$

$y = 3x + c$

sub (1; 12)

$12 = 3(1) + c$

$9 = c$

$\therefore y = 3x + 9$

$m = 3$

2) line b $\perp$ line a $\therefore m_1 \times m_2 = -1$

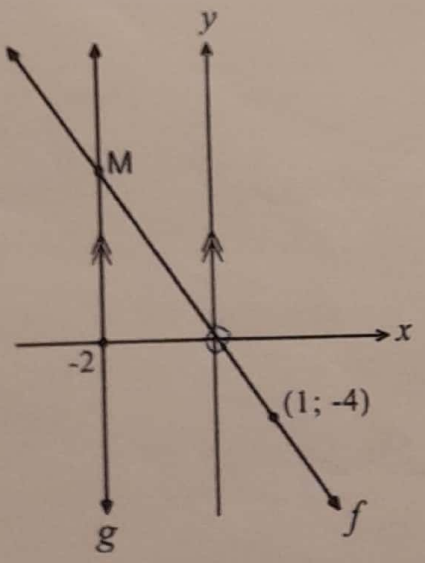
$3 \times m_2 = -1$

$m_2 = -\frac{1}{3}$

$y = -\frac{1}{3}x + c$ sub (0; -1) $\rightarrow$ c

$y = -\frac{1}{3}x - 1$

5.2



Use the information in the diagram above to find the following:

5.2.1 the equation of line *f* (3)

5.2.2 the equation of line *g* (2)

5.2.3 the coordinates of point M (2)

[13]

5.2.1 line f

$c = 0$ & sub (1; -4)

$y = mx + 0$

$-4 = m(1)$

$-4 = m$

$\therefore y = -4x + 0$

5.2.2 line g

$x = -2$ because // to y-axis

5.2.3 Point M

sub $x = -2$ into f

$y = -4(-2)$

$y = 8$

$\therefore M(-2; 8)$

Powered by CamScanner